

MADHAV RIMAL

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SUMMARY

Polyglot Software Engineer and AI enthusiast focused on building systems from first principles — from custom ML libraries and LLM orchestration frameworks to full-stack applications. Proven record in delivering AI applications for social good (Krishi-Chakshu) and full-stack platforms (Niyukt, Vrinda-cart), with an open-source contribution merged into the main branch of Vicharanshala (IIT Ropar). Strong foundation in applied mathematics and machine learning, with experience in NLP pipelines, computer vision, and end-to-end web development. Targeting the AI Core Basics Engineer / Full Stack Developer role at SAARC MASTS Tech Pvt. Ltd.

EDUCATION

Indian Institute of Technology Madras <i>BS Degree in Data Science and Applications (Ongoing)</i>	Chennai, Tamil Nadu 2023 – Present
Sanskrit Sahitya (Honours) <i>Study of knowledge structures, classical literature, and foundational pedagogy</i>	2023 – Present

EXPERIENCE

Your Role <i>Your Organization</i> - Description Point - 1 - Description Point - 2 - Description Point - 3	June 20XX – Present <i>Your City, State</i>
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PROJECTS

Open-Source Contribution — Vicharanshala (IIT Ropar) Python, NLP, Clustering - Contributed Question Clustering and Auto Smart Question Router feature; feature merged into main branch. - Implemented NLP-based clustering logic for improved discoverability.	2024
LLM Orchestration Platform for Mathematics Learning LangChain, RAG, Knowledge Graphs - Built a multi-agent LLM orchestration framework for personalised, video-based mathematics learning. - Implemented RAG for reference supply and designed a knowledge graph for adaptive topic delivery.	Aug 2024
Spam Email Detection System Python, NLP, Logistic Regression - Built a spam classifier using Bag-of-Words on ~100k emails; designed stacked logistic regression models. - Achieved ~90% accuracy on validation data.	Jan 2024
CrunchDAO Machine Learning Competition Python, Ensemble ML, Feature Engineering Ranked top ~300 out of ~10,000 global participants; self-designed complete ML pipeline.	May 2024
Niyukt Flask, SQL, Custom ORM, Async JS Built a centralised full-stack placement management system; developed a custom mini-ORM and responsive async dashboard.	2024
Vrinda-cart Node.js, JavaScript, MongoDB, REST API Full-stack e-commerce app; implemented end-to-end data flow using custom web components.	2024 – ongoing
Yantragya — Custom ML Library Python, NumPy, Mathematics Developing core ML algorithms from scratch in Python as a sandbox for understanding ML from first principles.	2024 – ongoing

ACHIEVEMENTS

- Open-source feature (Question Clustering & Smart Router) merged into main branch of Vicharanshala — IIT Ropar crowd-source project.
- Ranked top ~300 out of ~10,000 global participants in the CrunchDAO Machine Learning Competition (May 2024).
- Developed a functional spam detection system achieving ~90% accuracy as a first-year student.

TECHNICAL SKILLS

Languages: Python, JavaScript (Node.js / ES6+), Bash, SQL

AI / ML: Scikit-learn, TensorFlow, Keras, NumPy, Pandas, Matplotlib, Feature Engineering, NLP (BoW, Tokenization), Computer Vision (MobileNetV2)

LLM & AI Systems: LangChain, RAG (Retrieval Augmented Generation), Multi-Agent Orchestration, Knowledge Graphs, Prompt Engineering

Full-Stack & Web: Flask, Express.js, Node.js, HTML/CSS, REST API Design, Async JS / Fetch API, MongoDB (Mongoose), SQLite / MySQL / PostgreSQL

Developer Tools: Git, GitHub, Linux, VS Code

CO-CURRICULAR ACTIVITIES

IIT Madras BS Degree Program

2023 – Present

Active coursework in Machine Learning, AI Systems, Applied Mathematics, and NLP

Sanskrit Sahitya (Honours)

2023 – Present

Study of classical knowledge structures and pedagogical systems for integration with AI learning tools

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